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COMPUTER SCIENCE **9618/13**

Paper 1 Theory Fundamentals **October/November 2024**

1 hour 30 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS
- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

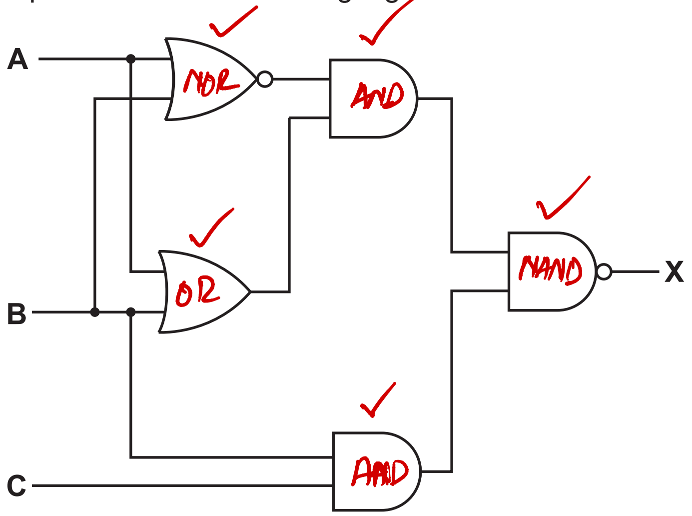
INFORMATION
- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].
- No marks will be awarded for using brand names of software packages or hardware.

This document has **20** pages. Any blank pages are indicated.

1 (a) Describe the operation of a 2-input XOR gate.

A 2-input XOR gate outputs 1 only when the two inputs are different, and outputs 0 when the two inputs are same. [1]

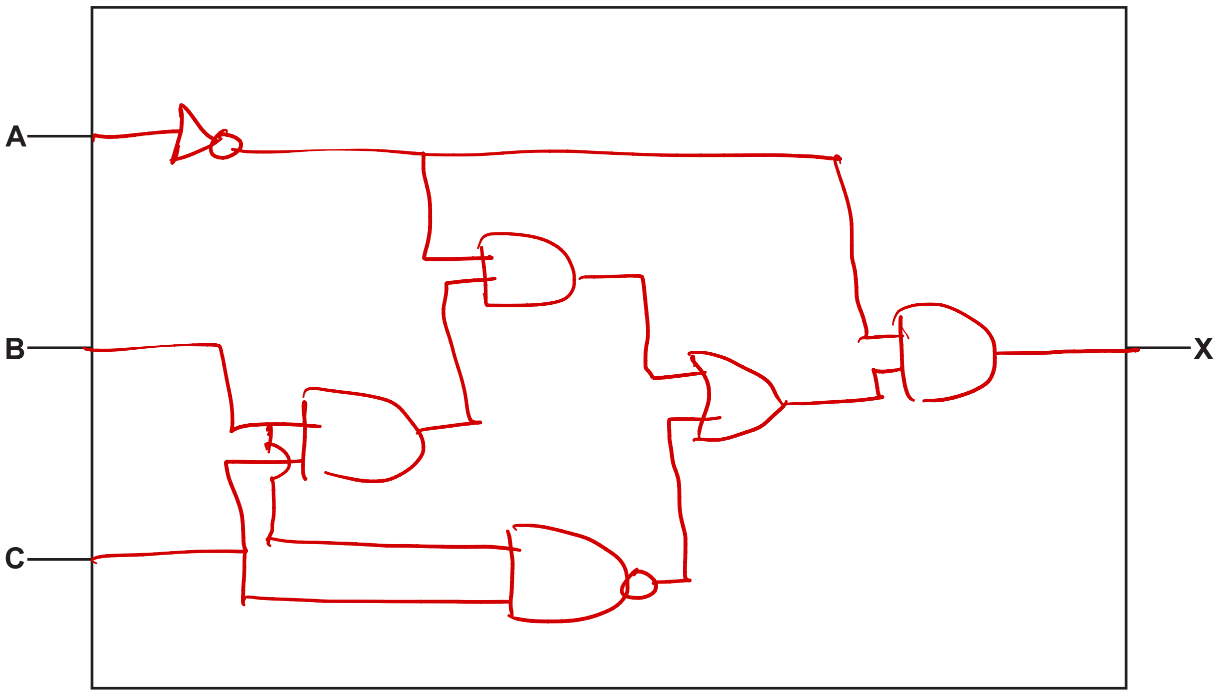
(b) Write the logic expression for the following logic circuit.



X = (((A NOR B) AND (A OR B)) NAND (B AND C)) [2]

(c) Draw the logic circuit for the logic expression:

X = ((NOT **A** AND (**B** AND **C**)) OR (**B** NAND **C**)) AND NOT **A**



[2]

2 A memory buffer uses Dynamic RAM (DRAM).

(a) Identify **two** differences between DRAM and Static RAM (SRAM).

Difference 1 DRAM needs to be constantly refreshed to retain data, whereas SRAM retains data without needing refreshing.

Difference 2 DRAM generally has higher storage capacity, while SRAM usually offers lower capacity per chip.

[2]

(b) Explain how a memory buffer is used when a computer is transferring data to its magnetic hard disk drive.

- When a comp. transfers data to HDD, a memory buffer is used to handle the difference in speed b/w the computer and the HDD.
- The CPU writes data quickly into buffer, allowing it to continue processing without waiting.
- Meanwhile, HDD reads stored data from buffer at its own slower speed.
- This prevents CPU being delayed by the HDD.

[4]

3 A car has an automatic braking system.

A sensor is used to measure the distance to objects that are in front of the car.

The car automatically brakes if an object is too close to the front of the car or the distance between the car and the object is decreasing too quickly. The closer the object is to the front of the car, the harder the car brakes so that the car slows down more quickly.

Explain the reasons why the automatic braking system of the car is a control system.

- System is a control system b/c it continuously takes input from distance sensor that measures how close objects are.
- Based on this input, the system automatically decides, whether bracking is needed ~~or~~ or not, how hard to brake.
- After braking, the new distance is measured again, and the braking is adjusted accordingly, forming a contineous feedback loop. [3]

4 A company makes ice cream and sells it to shops.

The ice cream is made in batches: a large quantity of one type and flavour of ice cream that is then split into smaller quantities for sale.

The company's owner has designed a relational database, `ICECREAM`, to store data about their ice cream and customers.

Some of the tables in the database are given. The database is not normalised.

`BATCH(`<u>`BatchID`</u>`, Type, Flavour, Size, SellingPrice, EndDate)`

`CUSTOMER(`<u>`CustomerID`</u>`, CompanyName, EmailAddress, TelephoneNumber)`

`SALE(`<u>`SaleID`</u>`, BatchID, CustomerID, Quantity, Date)`

(a) Identify **two** foreign keys in the table `SALE` **and** the table that each foreign key references.

Foreign key 1 BatchID

Table name 1 BATCH.

Foreign key 2 CustomerID

Table name 2 CUSTOMER.

[2]

(b) Write an SQL script to return the total quantity of ice cream sold to the customer with the ID of 0034E in the year 2023.

```
SELECT SUM(Quantity)
FROM SALE
WHERE CustomerID = "0034E" AND
      Date >= "01/01/2023" AND Date <= "31/12/2023";
```

[3]

(c) The table definition for `BATCH` is repeated here:

`BATCH(`<u>`BatchID`</u>`, Type, Flavour, Size, SellingPrice, EndDate)`

Sample data for the table `BATCH` is given:

BatchID	**Type**	**Flavour**	**Size**	**SellingPrice**	**EndDate**
KIV12	Plain	Vanilla	1	2.20	12/12/2024
RIC14	Plain	Chocolate	0.5	1.80	12/12/2024
TYL1	Non-dairy	Lemon	0.5	2.10	01/02/2025
FYV2	Non-dairy	Vanilla	0.25	1.50	02/02/2025
BIV13	Plain	Vanilla	1	2.20	02/02/2024

(i) Write an SQL script to define the table `BATCH`.

Include constraints (restrictions) on the data that can be entered into each field where appropriate.

```
CREATE TABLE BATCH(
    BatchID VARCHAR(6),
    Type VARCHAR(20),
    Flavour VARCHAR(20),
    Size REAL,
    SellingPrice REAL,
    EndDate DATE,
    PRIMARY KEY (BatchID)
);
```

[5]

(ii) The table BATCH is not normalised.

Normalise the database table BATCH.

Write the table definitions for your new tables.

Identify any primary and foreign keys in your tables.

Do **not** change or include the tables CUSTOMER and SALE.

```
CREATE TABLE ICECREAM(
    IceCreamID VARCHAR(6),
    Type VARCHAR(20),
    Flavour VARCHAR(20),
    Size REAL,
    SellingPrice REAL,
    PRIMARY KEY (IceCreamID)
);
```

```
CREATE TABLE BATCH(
    BatchID VARCHAR(6),
    IceCreamID VARCHAR(6),
    EndDate DATE,
    PRIMARY KEY (BatchID),
    FOREIGN KEY (IceCreamID)
    REFERENCES ICECREAM(IceCreamID)
);
```

[4]

(d) Complete the following table by defining each database term.

Database term	Definition
Entity	A real-world object or concept that is represented as a table in a DB.
Attribute	A single piece of information (field) that describes a property of an entity.

[2]

(e) A Database Management System (DBMS) supports data integrity.

Explain how a DBMS supports data integrity.

- A DBMS supports data integrity by enforcing rules such as PK, which ensures that each record is uniquely identified, and FK, which maintains correct links b/w related tables.
- It also applies refrential integrity constraints, so that updates or deletion in one table are correctly reflected in the related table.

[3]

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5 A computer programmer writes programs that are distributed with a software licence.

(a) Complete the table by writing the type of software licence each statement describes.

Statement	Software licence
A fee is charged for the software. The source code cannot be accessed. Users do **not** get a free trial.	Commercial licence, - Paid, closed-source, no trial.
Users can try the software before buying it. Users may **not** be able to access all features during the trial.	Shareware licence; - Trial version, pay for full access.
Software is usually free of charge. Users can access the source code and alter the program to their needs.	Opensource licence; Free, modifiable.

[3]

(b) Explain the reasons why it is important for the computer programmer to join a professional ethical body.

So a professional ethical body provides a code of ethics giving the computer programmer clear standards to follow when making decisions point number two being a member boost the programmer's reputation making clients and employers more confident in their skills and ethical behaviour point number three the body offers resources such as legal advice, training courses and carrier support to help programmers stay updated and protected professionally

[3]

DO NOT WRITE IN THIS MARGIN

6 A computer system stores text, images and sound.

(a) A character set is used to represent characters in a computer.

Identify **and** describe **one** character set.

Character set ASCII

Description Seven or eight bits per character, English alphabets and symbols. Stands for American standard code for information interchange. Enough for English letters numbers and basic symbols.

[2]

(b) The colour of each pixel in a bitmapped image is represented by 8 bits.

(i) State the largest number of different colours that can be represented by 8 bits.

$2^8 = 256$ Colours.

[1]

(ii) State **one** drawback of increasing the number of bits that represents each pixel in the bitmap image.

Drawback of increasing the number of bits that represent each pixel in the map image will increase increase the size of file.

[1]

(iii) A bitmap image can be compressed using lossy compression.

Explain the reasons why lossy compression is often suitable for a bitmap image.

Loss compression is suitable for bitmap images because it removes a small colour and detailed variations that are usually not noticeable to the human eye, allowing the image to still appear visually acceptable.

It also significantly reduces the file size, making it easier to store and transmit images efficiently.

[2]

(c) (i) Explain how an analogue sound wave is converted into digital data.

....

....

....

.... [2]

(ii) Describe **one** method of compressing a sound file using lossy compression.

One method is to remove frequencies that are outside the range of human hearing, because they do not contribute to what the listener can perceive, thereby reducing the file size.

....

....

.... [2]

The analog sound wave is sampled at regular time intervals, where the amplitude of the wave is measured at each point.

Each measured value is then quantised, meaning it is rounded to the nearest available level based on the number of bits used.

These quantised values are finally converted into binary form for digital storage and processing.

7 The following table shows part of the instruction set for a processor. The processor has two registers, the Accumulator (ACC) and the Index Register (IX).

Instruction		Explanation
Opcode	**Operand**	
`LDM`	`#n`	Immediate addressing. Load the number n to ACC
`LDD`	`<address>`	Direct addressing. Load the contents of the location at the given address to ACC
`LDI`	`<address>`	Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC
`LDX`	`<address>`	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC
`LDR`	`#n`	Immediate addressing. Load the number n to IX
`ADD`	`#n/Bn/&n`	Add the number n to the ACC
`ADD`	`<address>`	Add the contents of the given address to the ACC
`SUB`	`#n/Bn/&n`	Subtract the number n from the ACC
`SUB`	`<address>`	Subtract the contents of the given address from the ACC
`INC`	`<register>`	Add 1 to the contents of the register (ACC or IX)
`DEC`	`<register>`	Subtract 1 from the contents of the register (ACC or IX)

<address> can be an absolute or a symbolic address
# denotes a denary number, e.g. #123
B denotes a binary number, e.g. B01001010
& denotes a hexadecimal number, e.g. &4A

(a) The current contents of memory are shown:

Address	Data
50	54
51	55
52	50
53	52
54	100
55	25
56	50

The current contents of the ACC and IX are shown:

ACC	50
IX	45

Complete the table by writing the content of the ACC after each program has run.

	Instructions	ACC content

1	`LDD 50`
	`ADD #4`
	`ADD 54`
		54
		58
		158

2	`LDI 53`
	`DEC ACC`
	`ADD 56`
		50
		49
		99

3	`LDM #55` `SUB #5`	55 50

49
50
99

[3]

(b) The instruction set also includes these bit manipulation instructions:

Instruction		Explanation
Opcode	**Operand**	
AND	#n/Bn/&n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n/Bn/&n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n/Bn/&n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

Explain how bit manipulation can be used to clear the data in an 8-bit register.

Write the bit manipulation instruction that will be used.

Explanation All the bits will be reset to 0.

Instruction AND #0

[3]

8 (a) Convert the hexadecimal number 1FAB into denary.

Working .... 0001 1111 1010 1011 ....

(8107)₁₀

Denary value .... [1]

(b) Explain how to convert the two's complement integer 10011111 into denary. Give the denary value after conversion.

→ -ve

Explanation .... 10011111 .... -ve

(97) 01100001 +ve

-97.

1
32
64

Denary value ....

[3]

(c) Describe the difference between a right logical binary shift and a right arithmetic binary shift.

In a right logical binary shift, the bits are shifted to the right and the empty left most positions are filled with zeros.

In a right automatic binary shift, the bits are shifted to the right, but the most significant bit is copied into the left most position to preserve the sign of a number.

.... [2]

1 1 1 1 1 1 1 0 +   logical
0 0 1 1 1 1 1 1

1 1 1 1 1 1 0 +   -3
1 1 1 1 1 1 1 0   -2
1 1 1 1 1 1 1 1   -1

9 A computer is connected to a Local Area Network (LAN) that connects to a Wide Area Network (WAN).

(a) Describe the characteristics of a WAN.

....

....

....

.... [2]

(b) Copper cable can be used to transmit data in a network.

Complete the table by identifying **and** describing **two other** transmission media that can be used to transfer data in the WAN.

Transmission medium	**Description**
....	....
....	....

[4]

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(c) The computer on the LAN is used for real-time video conferences, where people connected to the internet communicate in real-time using video and audio.

Explain how bit streaming is used in a real-time video conference.

In real time video conference, bit streaming is used to send a continuous stream of video and audio data over the network as it is captured.

The data is compressed and transmitted in small packets and played back almost immediately on the receivers device, without waiting for the entire file to be downloaded.

This allows life communications with minimal delay between the sender and receiver.

[4]

(d) The LAN has a router. The router has a public IP address and a private IP address.

(i) State the purpose of a public IP address and a private IP address.

Public IP address Public IP address is used to uniquely identify the network or device on the Internet, allowing communication with external systems.

Private IP address Private IP address used within a local network to allow internal communication between devices, not directly accessible from the Internet.

[2]

(ii) The LAN uses subnetting.

Describe the purpose of subnetting in a network.

Subletting divides a larger network into smaller, more manageable sub networks.

It helps reduce network congestion by keeping local traffic within subnets and can improve security by isolating different groups of devices used by different categories of users.

[2]

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